

The Perceptron: Distribution-Free Learning and the Mistake Bound

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Abstract

Everything before this note assumed a conditional distribution and maximized its likelihood. The perceptron assumes *nothing*: no $p(x)$, no noise model, not even that examples are random — an adversary may choose them. What replaces the statistical guarantees is a *mistake bound*: on any stream separable with margin γ inside radius R , the perceptron makes at most $(R/\gamma)^2$ mistakes, ever. We state the algorithm (binary and multiclass), prove Novikoff's theorem in full, and demonstrate bound, convergence, and the non-separable failure mode in `perceptron.py`.

1 The online, distribution-free setting

Labels $y \in \{-1, +1\}$; examples (x_t, y_t) arrive one at a time, in any order, chosen by nature or an adversary. The learner maintains a linear predictor w and pays for every *mistake* ($y_t \neq \text{sign}(w^\top x_t)$). There is no distribution to be consistent for and no risk to converge to; the object of analysis is the cumulative mistake count. This is the learning-theory note's worst-case spirit taken all the way: not worst distribution — *worst sequence*.

2 The algorithm

Initialize $w_1 = 0$. At step t , predict $\hat{y}_t = \text{sign}(w_t^\top x_t)$; if $y_t w_t^\top x_t \leq 0$ (a mistake), update

$$w_{t+1} = w_t + y_t x_t, \tag{1}$$

otherwise leave w unchanged. The update is pure error correction: after it, $y_t w_{t+1}^\top x_t = y_t w_t^\top x_t + \|x_t\|^2$ — the offending example's margin moves up by $\|x_t\|^2$. (It is also SGD with step 1 on a perceptron loss $\max(0, -y w^\top x)$; the losses note places it in that family.)

3 The mistake bound

Theorem 1 (Novikoff). *Suppose some unit vector u separates the stream with margin γ : $y_t u^\top x_t \geq \gamma$ for all t , and $\|x_t\| \leq R$. Then the perceptron makes at most*

$$\boxed{M \leq \left(\frac{R}{\gamma}\right)^2} \tag{2}$$

mistakes on the stream, regardless of its length or order.

Proof. Only mistake steps change w ; let M be their number and consider w after the M -th mistake. Two quantities race.

Progress along u grows linearly. Each mistake adds $y_t u^\top x_t \geq \gamma$ to $u^\top w$:

$$u^\top w = \sum_{\text{mistakes}} y_t u^\top x_t \geq M\gamma.$$

The norm grows at most linearly. On a mistake, $y_t w^\top x_t \leq 0$, so

$$\|w + y_t x_t\|^2 = \|w\|^2 + 2y_t w^\top x_t + \|x_t\|^2 \leq \|w\|^2 + R^2,$$

hence $\|w\|^2 \leq MR^2$ after M mistakes.

Cauchy–Schwarz squeezes. Since $\|u\| = 1$,

$$M\gamma \leq u^\top w \leq \|w\| \leq \sqrt{M}R \implies \sqrt{M} \leq R/\gamma \implies M \leq (R/\gamma)^2. \quad \square$$

Read it: the bound involves *only geometry* — margin and radius — not the dimension, not the number of examples, not any probability. A million points or ten, random order or adversarial: at most $(R/\gamma)^2$ corrections. The script’s separable run makes a few dozen mistakes against a bound of a few hundred, then completes a clean pass.

Remark 1 (Non-separable data: the failure mode). *If no margin exists, the theorem’s premise fails and so does the algorithm’s behavior: updates never cease (the script’s noisy run shows mistakes accumulating linearly, forever). The perceptron has no notion of “good enough” — every mistake triggers the same full-strength correction, including on hopeless (mis-labeled) points. Handling non-separability by bounding the correction is exactly the Passive-Aggressive family; handling it by trading margin against slack is the SVM. Both next.*

4 Multiclass perceptron

One weight vector per class, prediction by best score: $\hat{y} = \arg \max_k w_k^\top x$. On a mistake, promote the truth and demote the pretender:

$$w_y \leftarrow w_y + x, \quad w_{\hat{y}} \leftarrow w_{\hat{y}} - x. \quad (3)$$

The binary case is recovered with $w = w_{+1} - w_{-1}$ (the two updates collapse into (1)), and a Novikoff-type bound holds with the multiclass margin $\gamma = \min_t (w_{y_t}^* - \max_{k \neq y_t} w_k^*)^\top x_t$ playing the same role. The same promote/demote shape is softmax regression’s gradient $(p - e_y) \otimes x$ hardened: probabilities replaced by a single all-or-nothing correction — the perceptron is the zero-temperature limit of the statistical model it abandoned.